

green product. Raw material should be selected such that it preserves the environment while meeting the product's requirements. Typically, this is crucial for product packaging materials like plastics, sheet metal and consumables. Also important are the product's shipping packaging materials like cartons, buffers and wooden crates.

Typically, attempt should be made to ensure that these materials are reusable or recyclable. Also, as there is little control once the product leaves the manufacturer, the raw materials shouldn't require any special process for recycling or reuse.

Design for cleaner production. This is typically a mandated process which most of the (contract) manufacturers follow. However, the onus is on the designers for selecting components that meet the process requirements of the manufacturer. One good example is the RoHS (Reduction of Hazardous Substances) directive, popularly known as lead-free manufacturing. Compliance with the ROHS directives requires the designers to select and use RoHS-compliant parts. It is applicable to all the parts used in a product and not restricted to electronic components. Additional requirement is implementing the design avoiding processes that are detrimental to the environment.

Design for energy efficiency. Energy efficiency has to be implemented both in hardware design as well as in software. In fact, an intelligent software can save energy in the most efficient way. Unless the architecture is selected for energy efficiency, it will be difficult to achieve the desired results.

Design for water conservation. It may be applicable to few products but is very important for medical devices and building automation systems which use plenty of water. One good example is the automatic water dispensers in toilet cleaners, which can be tuned to dispense water intelligently in order to conserve water.

Design for minimal consumption. This feature is very close to water conservation but with a little twist. It

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is predominantly implemented in software with the help of existing hardware features. One excellent example is the economy mode of a laser printer where the ink used is minimised, compromising the quality of print. For draft copies this is a very useful feature conserving energy. Another good example is the shutting down of the petrol engine of a hybrid car when the car is in idle mode. When the driver wants to move, the software starts the car with electrical drive and switches on the engine, thus saving precious petrol.

Design for service and repair. The design should be implemented such that the product can be diagnosed and the problem can either be solved by downloading the correct software or isolating the hardware module which has gone faulty so that the service engineer can go with the right part to solve the problem reducing multiple 'truck roles' at the client's place. In fact, the major victims of this kind of problem are printers and photocopiers, or typically products with high mechatronics content. Designing products with remote diagnostics capability reduces the number of visits of the service personnel to the site.

Design for reuse. This is very important if the product has to be sold in Asia. While reuse is very common in the automotive industry globally, reuse of electronic items like mobile phones and TVs is common in Asia. What this means is that the product has to meet global standards for environmental use as well as operating environments like mains voltage.

Design for remanufacture. This is essential for most of the test equipment and industrial (especially manufacturing) equipment due to the cost advantage over buying new equipment. In remanufacture, the system is com-

pletely dismantled and all the worn-out parts are replaced with new parts. The resulting system is available at a fractional cost of the new equipment. Unless the system is designed for it, remanufacture may not be possible. Typically, the designers have to identify the parts that can wear out very fast and ensure these can be replaced without much difficulty.

Design for disassembly. This typically means that at the end of its life, the system is taken apart into individual components (disassembly does not mean it can be reassembled) so that the components can be grouped and recycled or disposed of. So the designers have to envisage the disassembly process and select the component materials such that these can be grouped and recycled or disposed of safely.

Design for recycling. This step typically necessitates that the designers select parts which can ultimately be recycled to manufacture some other or the same product. This is true especially for metal and plastic parts, which take a lot of time to decompose when disposed of.

Design for safe disposal. Design engineers should avoid materials that will be difficult to dispose of (like plastic bags) or unsafe to dispose of (toxic chemicals and metals like mercury, cadmium and lead).

Design for efficient distribution. At the start of the design itself, the designers should understand how the product will be packaged and shipped to reach the end client. The idea behind this requirement is to reduce fuel consumption, the space occupied in the warehouse and loss due to damage (which results in material return and repair expenses). ●

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